Fiscal Year:	FY 2024	**Task Last Updated:**	FY 04/01/2024
PI Name:	Rithidech, Kanokporn Ph.D.		
Project Title:	Effects of Space Flights on the Proteome of Astronauts' Plasma		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Kanokporn.Rithidech@sbumed.org	**Fax:**	FY (631) 444-3424
PI Organization Type:	UNIVERSITY	**Phone:**	(631) 444-3446
Organization Name:	State University New York at Stony Brook		
PI Address 1:	Department of Pathology		
PI Address 2:	BHS T9 Health Sciences Center		
PI Web Page:			
City:	Stony Brook	**State:**	NY
Zip Code:	11794-8691	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-MIXEDTOPICS. Appendix E: Behavioral Health & Human Health Countermeasures Topics
Start Date:	04/01/2016	**End Date:**	12/31/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	Flight definition NOTE: End date changed to 12/31/2024 per NSSC information (Ed., 9/2/24). NOTE: End date changed to 05/31/2024 per T. Sirmons/JSC (Ed., 5/16/23). NOTE: End date changed to 12/31/2023 per PI and NSSC information (Ed., 3/21/23). NOTE: End date changed to 04/26/2023 per T. Sirmons/JSC (Ed., 3/3/23). NOTE: End date changed to 12/31/2022 per NSSC information (Ed., 1/4/22) NOTE: End date changed to 12/31/2021 per NSSC information (Ed., 12/31/20)		

	NOTE: End date changed to 12/31/2020 per NSSC information (Ed., 6/12/20) NOTE: End date changed to 3/31/2020 per NSSC information (Ed., 3/25/19)
Key Personnel Changes/Previous PI:	No change
COI Name (Institution):	Martin, Dwight Ph.D. (State University of New York, Stony Brook)
Grant/Contract No.:	NNX16AH80G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	NOTE: This is an integrated project consisting of Dr. Brian Crucian's "Functional Immune Alterations, Latent Herpesvirus Reactivation, Physiological Stress, and Clinical Incidence Onboard the International Space Station" directed research; and Dr. Richard Simpson's "The Impact of an ISS Mission on the Anti-Viral and Functional Properties of NK-cells, T-cells, B-cells and Dendritic Cells," Dr. Kanokporn Rithidech's "Effects of Space Flights on the Proteome of Astronauts' Plasma," and Dr. Honglu Wu's "DNA Damage in the ISS Astronaut's Lymphocytes and Their Association with Stress-Induced Immune Dysfunction" solicited research. Space flight results in exposure of astronauts to several stressors, such as space radiation, microgravity, and physiological stress, that could exacerbate the risks of adverse health effects. To protect astronauts, we must improve our understanding of molecular changes that influence immunological conditions associated with increased astronaut health risks. The in vivo response to the space environment is complex, involving multiple proteins associated with various signal transduction cascades, resulting in different outcomes. Molecular mechanisms responsible for such diverse consequences are poorly understood. It is, therefore, essential to characterize the protein signatures of responses to the space environment in blood plasma samples from astronauts, collected at pre-, in-, and post-flights. Such analyses should help to reveal a particular set of proteins causing adverse immunological changes and to develop methods that help to prevent, or at least to counteract, these effects. In this flight definition project, we will use cutting age proteomic technology to determine protein alterations, qualitatively and quantitatively, in plasma samples collected from astronauts before, during, and after space flights. Our findings will help to provide an understanding of the time course and etiology of immune changes induced by the space environment. Furthermore, since pre- and post-flight samples, in addition to the in-flight samples, will be evaluated in the same astronaut, the direct effects of the space environment can be determined. Hence, our findings will provide high-priority and highly relevant information to NASA. We will further correlate protein expression profiles to the available data on immune dysfunction detected in each astronaut. This approach makes it possible to determine potential predictive biomarkers for space-flight-induced immune dysregulation. Consequently, effective countermeasures against such harmful effects of the space environment can be identified.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our findings will deliver important information that should advance our understanding of the time course and etiology of immune changes induced by the space environment. Hence, our findings will provide high-priority and highly relevant information to NASA. Importantly, we will correlate protein expression profiles to the available data on immune dysfunction detected in each astronaut. This approach makes it possible to determine potential predictive biomarkers for space-flight-induced immune dysregulation. Such knowledge is important for the assessment of health risks and will facilitate the development of countermeasures that can help astronauts, space travelers, and people on Earth with the impairment of the immune system.
Task Progress:	We have completed a proteomic analysis and based on the results, we have prepared a manuscript entitled "Impact of the space environment and re-adaptation to Earth's gravity on the plasma proteome of astronauts." We are currently reviewing the manuscript internally and as soon as it's ready, we will submit it to the Journal of Proteomic Research for possible publication. Additionally, we will correlate our proteomic data with the available data on immune dysfunction detected in each astronaut by other groups of the core project "Functional Immune Study." This correlation will be a major task in 2024.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Articles in Peer-reviewed Journals	Martin D, Makedonas G, Crucian B, Peanlikhit T, Rithidech K. "The use of the Multidimensional Protein Identification Technology (MudPIT) to analyze plasma proteome of astronauts collected before, during, and after spaceflights." Acta Astronautica. 2022 Apr 1;193:9-19. https://doi.org/10.1016/j.actaastro.2021.12.054 , Apr-2022